Devil is in the Uniformity:Exploring Diverse Learners within Transformer for Image Restoration

Shihao Zhou[1,2] Dayu Li[1] Jinshan Pan[3] Juncheng Zhou[1] Jinglei Shi[1] Jufeng Yang[1,2]
1 VCIP&TMCC&DISSec,College of Computer Science,Nankai University
2 Nankai International Advanced Research Institute(SHENZHEN·FUTIAN)
3 School of Computer Science and Engineering,Nanjing University of Science and Technology
zhoushihao96@mail.nankai.edu.cn,ldy030911@gmail.com,sdluran@gmail.com
2112612@mail.nankai.edu.cn,jinglei.shi@nankai.edu.cn,yangjufeng@nankai.edu.cn

Abstract

Transformer-based approaches have gained significant attention in image restoration,where the core component,ie, Multi-Head Attention(MHA),plays a crucial role in capturing diverse features and recovering high-quality results. InMHA,heads perform attention calculation independently from uniform split subspaces,and a redundancy issue is triggered to hinder the model from achieving satisfactory outputs.In this paper,we propose to improve MHA by exploring diverse learners and introducing various interactions between heads,which results in a Hierarchical multI-head atteNtion driven Transformer model,termed HINT, for image restoration.HINT contains two modules,i.e.,the Hierarchical Multi-Head Attention(HMHA)and the Query-Key Cache Updating(QKCU)module,to address the redundancy problem that is rooted in vanilla MHA.Specifically,HMHA extracts diverse contextual features by employing heads to learn from subspaces of varying sizes and containing different information.Moreover;QKCU,comprising intra-and inter-layer schemes,further reduces the redundancy problem by facilitating enhanced interactions between attention heads within and across layers.Extensive experiments are conducted on12 benchmarks across 5 image restoration tasks,including low-light enhancement, dehazing,desnowing,denoising,and deraining,to demonstrate the superiority of HINT.The source code will be available at https://github.com/joshyZhou/HINT.

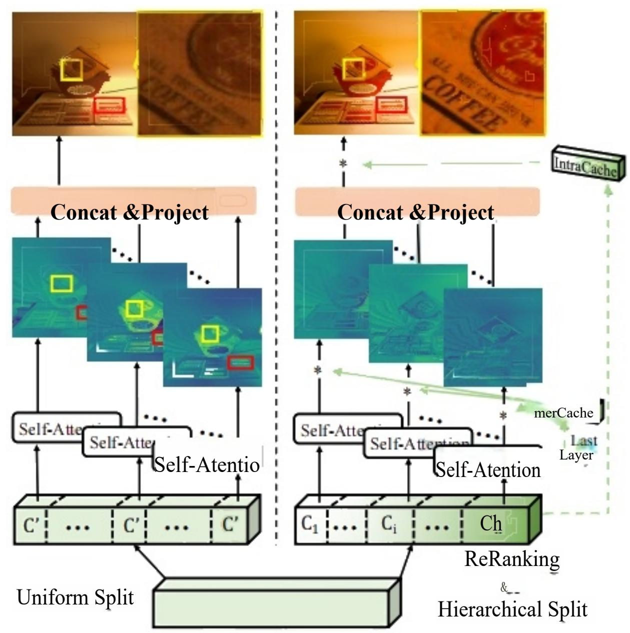


Figure 1.Comparisons between the vanilla MHA[2,37,82] (Lefi)and the proposed HMHA equipped with the QKCU module (Right),for the low-light enhancement task.The standard MHA assigns h heads with subspaces of the same size(C),and each head performs attention calculation independently.As a result, these heads intend to focus on the same regions (red boxes)and neglect the restoration of some degraded areas(yellow boxes),leading to an unsatisfactory output (losing details and introducing blur effect).In contrast,HMHA implements the reranking operation before a hierarchical subspace split,which encourages the model to learn diverse representative features.The QKCU enhances interactions between heads via intra-/inter-layer ways,modulating predicted features in HMHA and leading to better outputs.

1.Introduction

The Transformer-based frameworks [19,59]achieve promising performance in the field of image restoration. The success of this paradigm is rooted in the self-attention mechanism(SA)modeling non-local relations of pixels, which is crucial for recovering the global structure of the image.Many works pay attention to develop efficient SA variants for high-quality outputs [37,57,82].While it is worth noting that multi-head attention(MHA),employing multiple heads to perform self-attention computation in parallel from uniform split subspace,serves as the key fundamental component in embracing high computational efficiency and enhancing the diversity of the captured features.

“魔鬼在统一性中”:在Transformer中探索图像恢复的多样化学习者

世豪 Zhou[1]，2 大宇 Li[1] 金山 Pan[3] 君城 Zhou[1] 金雷 Shi[1] 聚峰 Yang[1]，2

1 VCIP & TMCC &DISSec，南开大学计算机科学学院

2 南开国际高等研究院（深圳· FUTIAN）

3 南京大学计算机科学与工程学院 zhoushihao96@mail.nankai.edu.cn，ldy030911@gmail.com，sdluran@gmail.com

2112612@mail.nankai.edu.cn,jinglei.shi@nankai.edu.cn,yangjufeng@nankai.edu.cn

摘要

基于Transformer的方法在图像修复领域备受关注，其中核心组件——多头注意力机制（MHA）在捕捉多样化特征和恢复高质量结果方面发挥着关键作用。在MHA中，各头独立于均匀分割子空间进行注意力计算， 由此引发冗余问题，阻碍模型获得满意 输出。本文提出通过探索多样化学习器并引入头间多种交互来改进MHA，最终构建出用于图像修复的分层多头注意力驱动Transformer模型HINT。HINT包含两个模块：分层多头注意力（HMHA）和查询 关键字缓存更新 （QKCU）模块 ， 旨在解决传统MHA中根植的冗余问题 。 具体而言 ， HMHA 通过利用不同规模且包含多样化信息的子空间进行学习，从而提取多样化的上下文特征。此外； QKCU，包含层内和层间方案，通过促进层内和跨层注意力头之间的增强交互，进一步减少了冗余问题。我们在5个图像恢复任务的12个基准上进行了扩展实验，包括弱光增强、去雾、去雪、去噪、去雨，以证明HINT的优越性。源代码将 在 https://github.com/joshyZhou/HINT。

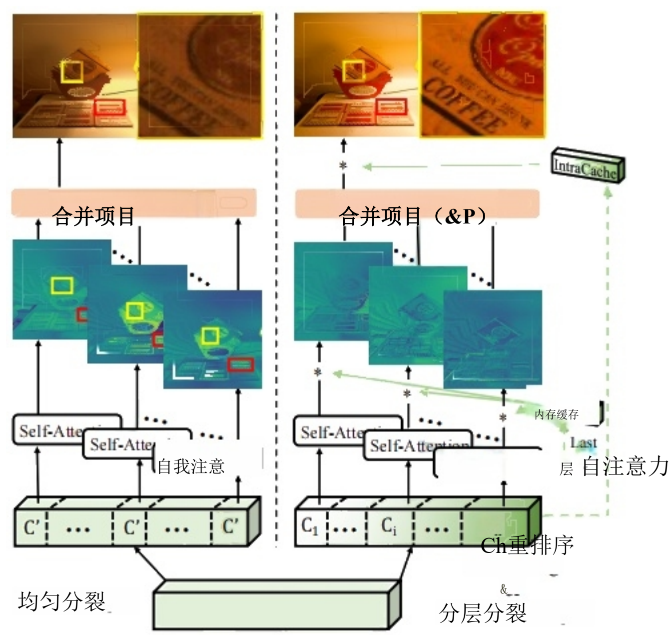


图 1. 原始MHA[2,37,82]（Lefi） 与配备 QKCU 模块的改进 HMHA（右图）在弱光增强任务中的对比。标准MHA将h个头分配相同大小的子空间(C)，各头独立进行注意力计算。结果导致这些头集中关注相同区域（红色框），而忽略部分退化区域的修复（黄色框），从而产生细节丢失和模糊效果的不理想输出。相比之下， HMHA 在分层子空间分割前实施重排序操作，促使模型学习多样化代表性特征。 QKCU 通过层内/层间方式增强头间交互，调控 HMHA 中预测特征，最终获得更优输出。

1. 引言

基于Transformer的框架[19,59]在图像修复领域取得了令人瞩目的性能。该范式的成功源于自注意力机制（SA）对像素非局部关系的建模，这对于恢复 图像 的全局结构至关重要。 许多研究致力于开发高效的SA用于高质量输出的变体[37,57,82]。值得注意的是，多头注意力机制（MHA）通过 多个头并行 - 从均匀分割子空间中进行自 - 适应性注意力计算，是实现高效计算和增强特征多样性 的关键基础组件。

The conventional MHA mechanism has a drawback of the redundancy issue.Some researches [48,49,64],in the field of NLP,have pointed out that specialized heads contribute most to the final decision while others can be pruned. In this paper,we demonstrate that this problem exists in the restoration tasks,and further find out that the redundancy issue can be traced to the focus on the same region by different heads.As shown in Figure 1,the standard MHA mechanism assigns h heads with subspaces of the same dimension(C'),where each head performs attention calculations in parallel and independently of the others.The visualized features from different heads disclose the problem of the vanlla MHA-that is,the inclination to pay attention to the same region (red boxes)which is redundant,and neglect of restoration on some degraded areas (yellow boxes)which causes unpleasant outputs.

To address this limitation,we improve the MHA from two perspectives. **First,** the heads learn from subspaces of uniform size,which contain similar information,leading to the problem of focusing on the same regions.Intuitively, to mitigate this issue,we introduce a ranking paradigm in terms of channel similarity before a hierarchical subspace splitting.In this way,each subspace contains information independent from others,and their sizes are different.This design explores heads as diverse learners,resulting in a hierarchical Multi-Head Attention module,namely HMHA, to replace the vanilla counterpart.It is capable of extracting diverse representative features,compared to the conventional MHA.**Second,** a lack of collaboration between heads makes the redundancy problem worse.We propose to enhance interactions between heads in intra-and inter-layer schemes through a Query-Key Cache Updating(QKCU) mechanism.Specifically,the intra-layer cache serves as a gating module,enhancing useful information in aggregated features that are captured by heads.On the other hand,the inter-layer cache emphasizes modulating the calculated attention score of each head,using history attention scores.Both intra-and inter-modulation are dependent on inputs,which improve the capability of HMHA to learn diverse contextual representations.Building upon the two key components,i.e.,HMHA and QKCU,we propose HINT,a Hierarchical multi-head atteNtion driven Transformer model for image restoration.

Overall,we summarize the contributions of this work:

- We present HINT,a Hierarchical multi-head atteNtion driven Transformer model for removing undesired degradations from images.HINT demonstrates the effect of exploring diverse learners in Multi-Head Attention(MHA) and enhancing interactions via inter-and intra-layer ways for the problem of image restoration.
- HINT introduces Hierarchical Multi-Head Attention (HMHA),which alleviates the redundancy issue in standard MHA by enabling the model to learn distinctive contextual features from different subspaces.Additionally,HINT incorporates the Query-Key Cache Updating(QKCU)mechanism,combining intra-and inter-layer schemes to enhance interaction between heads.
- Extensive qualitative and quantitative evaluations are conducted on 12 benchmarks for 5 typical image restoration tasks:low-light enhancement,dehazing,desnowing,denoising,and deraining,where HINT performs favorably against state-of-the-art algorithms in terms of restored image quality and model complexity.

2.Related Work

2.1.Image Restoration

Capturing images in unsatisfactory environments often results in low-quality outputs,negatively impacting downstream tasks [5,20,28].Image restoration offers a plausible solution by recovering clear images from undesired degradations,e.g.,haze [25,32,56],rain streaks [22,51,66],and low-light [2,70,75].Over the past decades,the research community has witnessed the paradigm shift from conventional hand-crafted approaches [9,21]to learning-based CNN models [38,45,79]. To achieve improved restoration performance,various efficient modules and advanced architecture designs have been proposed.Among these,residual feature learning[23,40,86]with skip connection,encoder-decoder architecture [10,30,80]for hierarchical representations,and attention mechanisms[17,55,83]to emphasize important signals have become the popular ingredients

In recent years,Transformer-based models [59]have been adapted for low-level tasks,achieving significant advancements across various image restoration tasks [2,57, 74].IPT [3]is a pioneering work that applies the vision Transformer [19]to low-level tasks,and obtains surprising results.The quadratic complexity of vanilla self-attention hinders it from applying high-resolution inputs,prompting researchers to explore solutions for reducing computational loads.To address this,Restormer [82]computes attention scores along channel dimension.Another approach is window-based attention [43],which is adopted by models such as Uformer [69]and SwinIR [37]. Although these attention mechanisms successfully alleviate the computational burden,they still rely on the vanilla MHA[59],which incurs redundancy issue [47,49,73]and further limits the representation capacity of the model.

2.2.Multi-Head Attention

As a fundamental component of the Transformer,MHA plays a crucial role in capturing diverse relationships and achieving impressive performance in practice.Unfortunately,it is well known in the research community that not all heads contribute equally to the given task [60].Previous works have attempted to address this by incorporating

传统多头注意力机制存在冗余问题的缺陷。自然语言处理领域的研究[48,49,64]指出，虽然专业头对最终决策贡献最大，但其他头可以被剪枝。本文证明该问题同样存在于图像修复任务中，并进一步发现冗余问题源于不同头对同一区域的过度关注。如图1所示，标准多头注意力机制将h个头分配到相同维度的子空间（C '），每个头独立并行执行注意力计算。不同头的可视化特征揭示了传统多头注意力机制的缺陷：过度关注同一区域（红色方框）导致冗余，而忽视某些退化区域的修复（黄色方框）则产生不良输出。

为解决这一局限性，我们从两个维度改进了多头注意力机制（MHA）。**首先，** 头部网络从统一尺寸的子空间中学习，这些子空间包含相似信息，容易导致注意力过度聚焦同一区域。为缓解这一问题，我们创新性地在层级化子空间分割前引入通道相似性排序机制。通过这种方式，每个子空间包含独立信息且尺寸各异。该设计将头部网络视为多元学习者，最终形成层级化多头注意力模块 HMHA，取代传统版本。相较于常规MHA，该模块能提取更多样化的代表性特征。**其次，** 头部网络间协作不足会加剧冗余问题。我们提出通过查询键缓存更新（QKCU）机制增强层内与层间交互。具体而言，层内缓存作为门控模块，能强化头部网络捕获的聚合特征中的有效信息。另一方面，层间缓存侧重于通过历史注意力分数来调节每个头的计算注意力分数。这种内外调节机制均依赖输入数据，从而提升 HMHA 学习多样化上下文表征的能力。基于 HMHA 和 QKCU 这两个核心组件，我们提出HINT（层次化多头注意力驱动）

图像恢复的变压器模型。

总体而言，我们总结了本研究的贡献：

- 我们提出了一种用于去除图像中不期望退化的HINT模型，该模型采用层次化多头注意力驱动的Transformer架构。HINT通过探索多头注意力（MHA）中多样化的学习器，并通过层间与层内交互增强，展示了其在图像修复问题上的效果。
- HINT引入了分层多头注意力机制（HMHA），通过使模型能够学习独特特征，从而缓解了标准多头注意力机制（MHA）中的冗余问题。
- 来自不同子空间的上下文特征。此外，HINT还引入了查询键缓存更新（QKCU）机制，通过结合层内与层间方案来增强网络头之间的交互。
- 针对5项典型图像修复任务（包括弱光增强、去雾、去雪、去噪及去雨），HINT算法对12个基准测试指标进行了全面的定性与定量评估。结果显示，该算法在图像质量恢复效果和模型复杂度方面均优于当前最先进的算法。

2. 相关 工作

2.1. 图像 恢复

在不理想环境中拍摄图像往往会导致低质量输出，对下游任务产生负面影响[5,20,28]。图像修复通过从不希望的退化现象（如雾霾[25,32,56]、雨痕[22,51,66]和弱光[2,70,75]）中恢复清晰图像，提供了一种可行的解决方案。过去几十年间，研究界见证了从传统手工方法[9,21]到基于学习的CNN模型[38,45,79]的范式转变。为提升修复性能，已提出多种高效模块和先进架构设计。其中，采用跳跃连接的残差特征学习[23,40,86]、用于分层表征的编码器-解码器架构[10,30,80]以及强调重要信号的注意力机制[17,55,83]已成为主流方案。

近年来，基于Transformer的模型[59]已被成功应用于低级任务，在各类图像修复任务中取得了显著进展[2,57，74]。 IPT [3]作为开创性研究，将视觉Transformer[19]应用于低级任务并取得突破性成果。传统自注意力机制的二次复杂度限制了其对高分辨率输入的处理能力，促使研究者探索降低计算负担的解决方案。为此，Restormer[82]沿通道维度计算注意力分数。另一种方法是窗口注意力机制[43]， 如Uformer [69]和SwinIR[37]等模型均采用该方案。尽管这些注意力机制有效缓解了计算压力，但它们仍依赖于基础的多头注意力机制[59]，该机制存在冗余问题[47,49, 73]，进一步限制了模型的表征能力。

2.2. 多头注意力机制

作为Transformer的核心组件，多头注意力机制（MHA）在捕捉多样化关系和实现卓越性能方面发挥着关键作用。然而研究界普遍认识到，并非所有头结构对特定任务的贡献度均等[60]。现有研究尝试通过整合

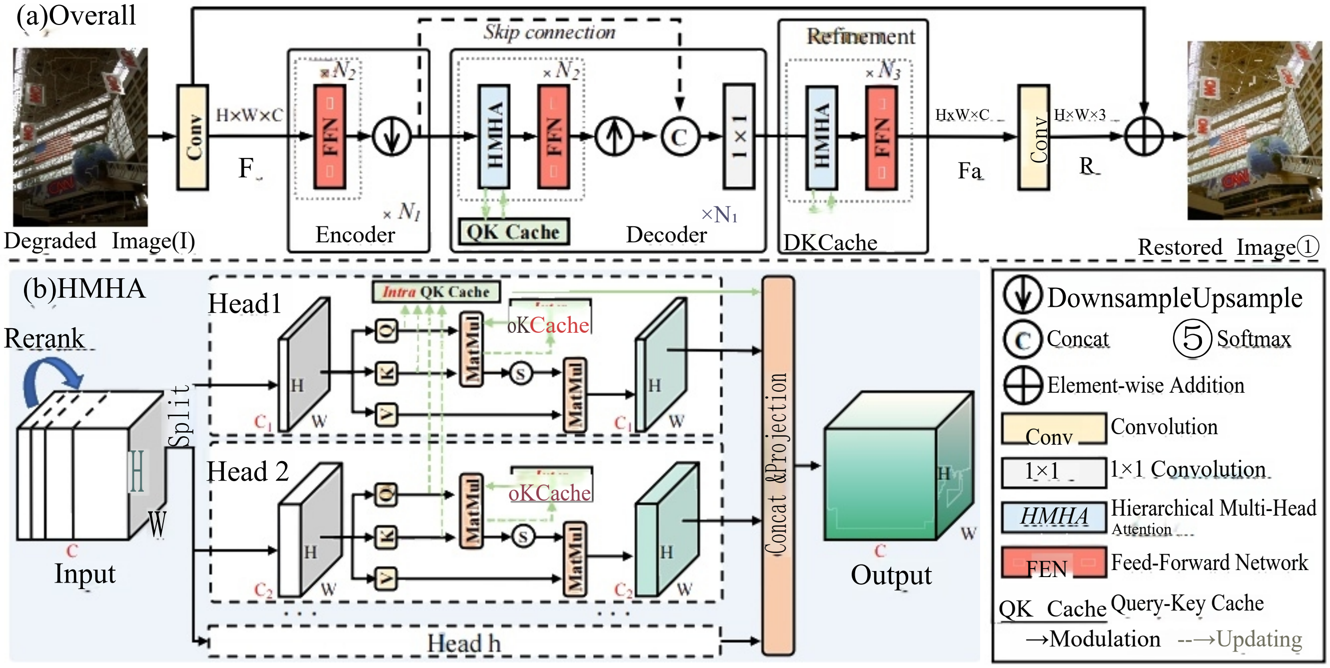


Figure 2.Ilustration of the proposed Hierarchical multi-head atteNtion driven Transformer model (HINT).(a)Overview architecture of the proposed HINT.(b)Hierarchical Multi-Head Attention (HMHA)mechanism.

interaction or collaboration between heads[1,35,85],however,the expressive power gained from these methods remains limited,as the heads still operate independently [73]. Another potential remedy is to modulate the query,key, and value projections of heads [12,41].While this modifies the underlying information flow,it lacks adaptability to inputs due to its static nature [73].More recently,the community has explored dynamically modulating attention scores [48,54,64],which ensures improvement of expressive power without introducing much computational loads.

In this study,towards solving the image restoration problem,we build upon the basic idea of modulating attention scores [48,73],while further mitigating the limited express power issue that is rooted in the vanilla MHA.Specifically, standard MHA assigns each head with subspaces using the same dimension size [3,37,82],limiting the ability of the heads to learn distinctive features and resulting in redundancy.In contrast,we propose to learn hierarchical representations using HMHA that extracts diverse contextual information from heads using different dimensional subspaces.Additionally,we introduce the QKCU mechanism, incorporating both intra-and inter-layer schemes,to enhance interaction among heads.Unlike existing restoration methods to implement modulation of attention scores in a static projection way [8,88],HINT makes the modulation dynamic,achieving better model expressiveness.

3.Method

Overall Pipeline. As illustrated in Figure 2,the proposed HINT follows an encoder-decoder architecture.Given a degraded input $I\in R^{H\times W\times 3}$,a convolutional layer is first used to extract the shallow feature $Fs\in R^{H\times W\times C}$,where H,W,and C represent the height,width and channel dimension,respectively.The shallow feature is then processed through the N_1-level restoration pipeline to produce a deep feature $Fa\in R^{H\times W\times C}$.At each level of both the encoder and decoder,there are N_2basic blocks,along with a convolution layer for down-sampling or up-sampling.Following prior works [29,88,89],we adopt an asymmetric design,which omits the self-attention mechanism in the encoder part,to boost the restoration performance.Thus,the basic block in the encoder consists only of a Feed-Forward Network(FFN)[82],while in the decoder,the block includes both the proposed HMHA and an FFN.We employ skip connection operation with 1×1 convolution to take advantage of features from the encoder in the decoder layers.After that,a refinement stage,consisting of N_3basic blocks,is developed to further enhance the learned features. Finally,a3×3 convolution layer processes the deep feature Fa to generate the residual image $R\in R^{H\times W\times 3}$.The restored image is obtained by adding the residual image to the degraded one,i.e.,I=I+R.

3.1.Hierarchical Multi-head Attention

Standard MHA assigns each head with a subspace of the same size containing similar information,which hampers the ability to learn distinctive features,leading to redundancy.To address this,HINT forms the HMHA to explore different dimensional subspaces,encouraging each head to learn diverse contextual information.

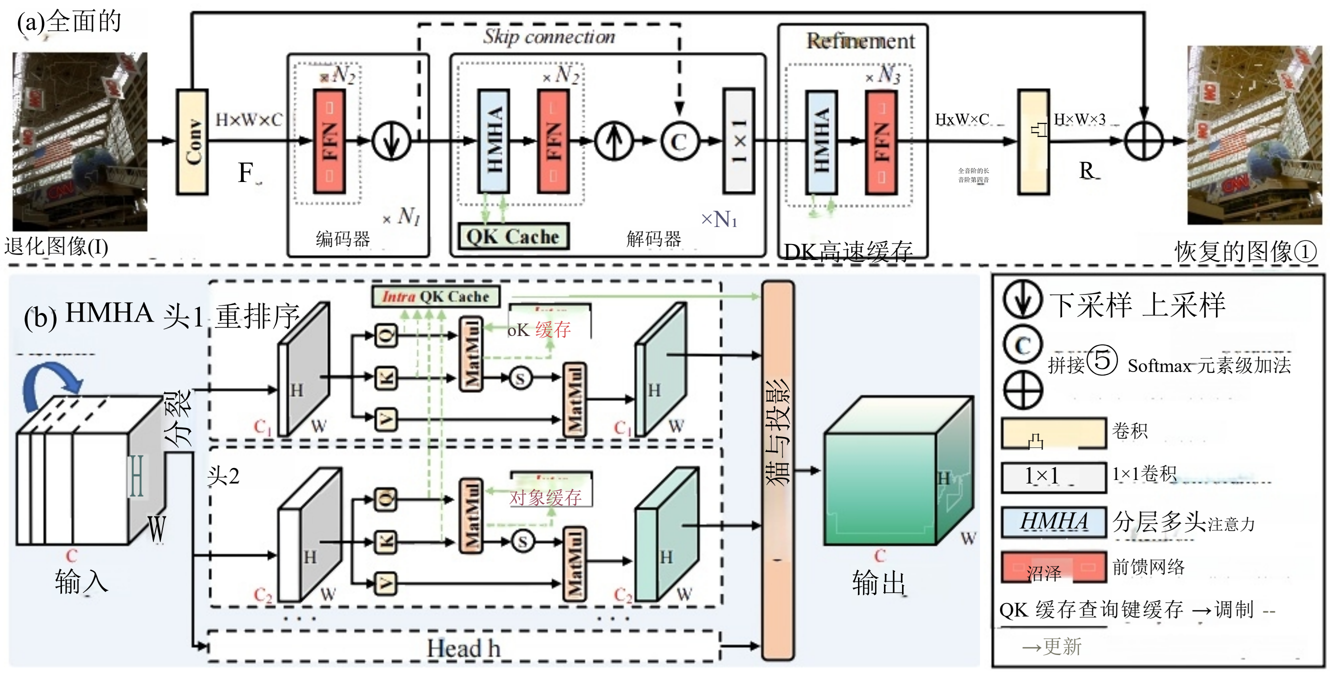


图2.所提出的分层多头注意力驱动Transformer模型（HINT）示意图。(a)HINT模型的总体架构。(b)分层多头注意力（HMHA）机制。

虽然头层[1,35,85]之间存在交互或协作，但这些方法获得的表达能力仍然有限，因为头层仍保持独立运作[73]。另一种潜在解决方案是调节头层的查询、键和值投影[12,41]。尽管这改变了底层信息流，但由于其静态特性[73]，这种方法缺乏对输入的适应性。最近，

研究者已探索动态调节注意力分数的方法[48,54,64]，该方法可在不引入过多计算负荷的情况下确保表达能力的提升。

在本研究中，为解决图像修复问题，我们基于注意力分数调制的基本思想[48,73]，同时进一步缓解了原始MHA模型中固有的表达能力不足问题。具体而言，标准MHA模型使用 相同维度的子空间[3,37,82]为每个注意力头分配空间，这限制了头部学习独特特征的能力并导致冗余。相比之下，我们提出通过 HMHA 学习层次化表征，该方法利用不同维度的子空间从各注意力头中提取多样化的上下文信息。此外，我们引入 QKCU 机制，结合层内与层间方案以增强头部间的交互作用。与现有修复方法通过静态投影方式调制注意力分数[8,88]不同，HINT实现了动态调制，从而获得更强的模型表达能力。

3. 方法

整体流程。 如图2所示，所提出的HINT采用编码器-解码器架构。给定一个解-

输入图像$E\in RH\times W\times 3$首先通过卷积层提取浅层特征$Fs\in RH\times W\times C$（其中H、W和C分别代表图像的高度、宽度和通道维度）。随后，浅层特征通过 N_l 级恢复流程处理生成深层特征$Fa\in RH\times W\times C$。在编码器和解码器的每一层级中，均包含N_2basic 个卷积块及一个用于下采样或上采样的卷积层。与现有研究[29,88,89]类似，我们采用非对称设计方案，省略编码器中的自注意力机制以提升恢复性能。因此，编码器的基本单元仅包含前向神经网络（FFN）[82]，而解码器则同时包含所提出的 HMHA 和 FFN 。我们采用1×1卷积的跳跃连接操作，以便在解码层中利用编码器的特征。随后，通过由N_3basic 个卷积块组成的精炼阶段进一步增强学习特征。最终，3×3卷积层对深层特征Fa进行处理生成残差图像$R\in RH\times W\times 3$。恢复图像通过将残差图像与降质图像相加获得。一、即I=I+R。

3.1. 层次化多头注意力机制

标准MHA为每个头分配一个相同大小的子空间，其中包含相似信息，这会削弱学习独特特征的能力，导致冗余。为解决这一问题，HINT构建了探索不同维度子空间的 HMHA，鼓励每个头学习多样化的上下文信息。

We begin by revisiting the scaled dot-product attention mechanism in MHA[59],which is adopted in mainstream approaches [37,82,88].Given a normalized input tensor $X \in RH\times W\times C$,the attention score is calculated as:

$$\text{Attention(Q,K,V)} = \text{Softmax}\left(\frac{\mathbf{QK}^T}{\sqrt{d_k}}\right)\mathbf{V}, \quad (1)$$
$$Q=XWQ,K=XWK,V=XWv,$$

where $Wo\in RC\times dk$,$Wk\in RC\times dk$,and $Wv\in RC\times d$ are the linear projection matrices for the query(Q),key (K),and value(V).Notably,for self-attention,the dimensions of key and value are equal,i.e.,dk=dv.

The conventional MHA [19,82]leverages multiple heads to perform scaled dot-product attention in parallel. By operating the attention function on different subspaces, the representation power of the model is supposed to be enhanced.Specifically,standard MHA adopts h heads to learn representations of(Q,K,V)in different subspace,i.e, dk/h.The MHA can be formally expressed as:

$$\text{MultiHead(X)=Concat}(H_1,H_2,\ldots,H_2)Wp, \quad (2)$$
$$H_i = \text{Attention}(\mathbf{XW}_Q^i, \mathbf{XW}_K^i, \mathbf{XW}_V^i),$$

$\mathbf{W}_Q^i \in \mathbb{R}^{C\times d_k/h}, \mathbf{W}_K^i \in \mathbb{R}^{C\times d_k/h}$,and $W\in RC\times do/h$ are the projection matrices for the i-th head.The output projection matrix $Wp\in Rdo\times dout$ aggregates the features captured by all heads.

To enhance the express power of MHA,we introduce a hierarchical representation learning process.To be specific,the proposed HMHA assign each head to a different dimensional subspace by splitting the channel space as $C=[C1,C_2,\ldots,Ch]$with $C_1\leq C_2\leq\ldots\leq Ch$.Before performing this split,we first rerank the channels based on their similarity to ensure that each head focuses on distinct semantic features.Each head then operates within its own subspace,performing dot-product attention in different subspace to capture diverse contextual information.

3.2.Query-Key Cache Updating Mechanism

The HMHA mechanism encourages multiple heads to learn features containing diverse contextual information, while these heads still work independently as conventional works [37,82].In this way,there is a lack of collaboration within the layer and across the entire model.As a result, the model is hindered from achieving optimal restoration performance.To mitigate this issue,we develop a QKCU mechanism that enhances interaction among heads,both within layers and across layers,as illustrated in Figure 3.

Intra Query-Key Cache Modulation &Updating.Given two input tensors,including the IntraCache $Fintra \in RC\times HW$ and Output of HMHA $Fout \in RC\times HW$,we begin by computing the sum of the two features $\mathbf{F}^s_{\text{intra}} = Fintra+Fout$.To selectively retain the most informative elements within the flow ofinformation,we introduce a gating mechanism,which can be formulated as:

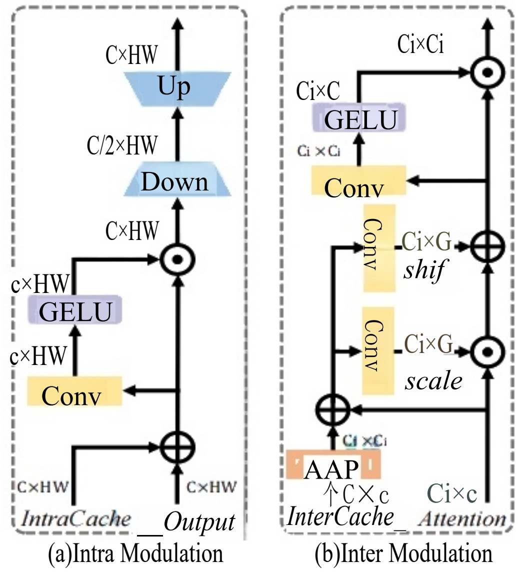


Figure 3.Query-Key Cache Updating Mecanism.

$$\mathbf{F}_{\text{gated}} = \text{GELU}(\text{Conv}(\mathbf{F}^s_{\text{intra}})) \odot \mathbf{F}^s_{\text{intra}}, \quad (3)$$

where $Fgated\in RC\times HW$ is the result of the gating mechanism,◎ denotes element-wise multiplication,Conv(·)and GELU(·)are convolution operation and the GELU activation function [26],respectively.Then,we adapt this transformed feature via feature compression and rebuilding:

$$\mathbf{F}^o_{\text{Intra}} = \text{Conv}_{\text{up}}(\text{Conv}_{\text{down}}(\mathbf{F}_{\text{gated}})), \quad (4)$$

where $\mathbf{F}^o_{\text{Intra}} \in \mathbb{R}^{C\times HW}$ is the output feature after the modulation process,Convup(·)and Convdown(·)are convolution operations that project the channel dimension into higher and lower levels,respectively.This design encourages the model to focus on the key features of the data.

Next,to update the Intra Query-Key Cache in each layer, we compute the sum of the Query(Q)and Key(K)projections for each head as follows:

$$F^2=Q^2+Ki, \quad (5)$$
$$Fintra=\text{Concat}(F^1,\ldots,Fh)$$

Inter Query-Key Cache Modulation &Updating.Given two input tensors,the InterCache $Finter\in RC'\times C"$and the dot-product attention of query and key $Fatt\in RCi\times C;$, we first resize Finter to obtain Finter $\in RC:\times C:$.After that,we compute the modulation components by summing Fatt and the resized feature map Finter,resulting in: $\mathbf{F}^s_{\text{inter}} = \mathbf{F}_{\text{att}} + \hat{\mathbf{F}}_{\text{inter}}$.We then calculate 'scale'and 'shift'components for pixel-wise modulation:

$$\mathbf{F}_{\text{shift}} = \mathbf{F}^s_{\text{inter}}\mathbf{W}\text{shift}, \mathbf{F}_{\text{scale}} = \mathbf{F}^s_{\text{inter}}\mathbf{W}_{\text{scale}}, \quad (6)$$
$$Fm=Fscale \circledcirc Fatt+Fshift,$$

我们首先回顾MHA[59]中提出的缩放点积注意力机制，该机制被主流方法[37,82,88]所采用。给定归一化输入张量X∈RH×W×C，注意力分数的计算公式为：

$$\text{Attention(Q,K,V)} = \text{Softmax}\left(\frac{\mathbf{QK}^T}{\sqrt{d_k}}\right)\mathbf{V}, \quad (1)$$

Q=XWQ,K=XWK,V=XWv,

其中Wo∈RC×dk、Wk∈RC×dk、Wv∈RC×d分别为查询(Q)、键(K)和值(V)的线性投影矩阵。值得注意的是，在自注意力机制中，键与值的维度相等，即dk=dv。

传统多头注意力机制（MHA）[19,82]通过并行计算多个头来实现缩放点积注意力。该机制通过在不同子空间上运行注意力函数，旨在增强模型的表征能力。具体而言，标准MHA采用h个头分别学习不同子空间（Q、K、V）的表征，即dk/h个头。其数学表达式可表述为：

多头网络(X)= H_1 与 H_2，...，H_2 的拼接Wp (2)

$$\text{H}_\text{i} = \text{Attention}(\mathbf{XW}^i_Q, \mathbf{XW}^i_K, \mathbf{XW}^i_V),$$

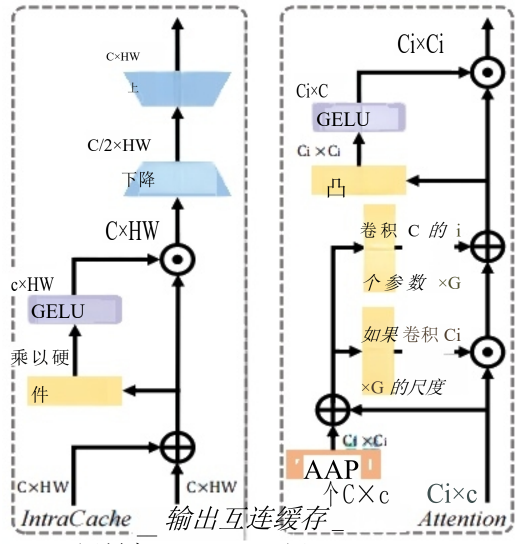
$\mathbf{W}^i_Q \in \mathbb{R}^{C\times d_k/h}, \mathbf{W}^i_K \in \mathbb{R}^{C\times d_k/h}$和 W∈RC×do/h是第i个头的投影矩阵。输出投影矩阵Wp∈Rdo×dout汇总了所有头捕获的特征。

为增强MHA的表达能力，我们引入了一种分层表征学习方法。具体而言，该 HMHA 通过将通道空间划分为C=[C1，C_2，....，Ch]，使每个头被分配到不同的维度子空间。 $C_1 \le C_2 \le ... \le$Ch. 之前在进行这种分割时，我们首先根据通道间的相似性对它们进行重新排序，以确保每个头专注于不同的语义特征。随后，每个头在其各自的子空间内运作，通过在不同子空间中执行点积注意力来捕捉多样化的上下文信息。

3.2 查询键缓存更新机制

HMHA 机制鼓励多个头学习包含多样化上下文信息的特征，同时这些头仍像传统方法[37,82]那样独立运作。这种方式导致层内及整个模型缺乏协作，因而阻碍模型达到最优恢复性能。为解决此问题，我们开发了一种 QKCU
如图3所示，该机制可增强头部之间的相互作用，包括层内及跨层的相互作用。

查询键内缓存调制与更新。给定两个输入张量，包括 IntraCacheFintra∈RC×HW 和 HMHA 输出 Fout∈RC×HW，我们首先计算这两个特征的和$\mathbf{F}^{s}_{\text{intra}} =$



(a)调制内 (b)互调

图3. 查询键缓存更新机制。

筛选+过滤。为选择性保留信息流中最富信息量的元素，我们引入了一种门控机制，其可表述为：

$$\mathbf{F}_{\text{gated}} = \text{GELU}(\text{Conv}(\mathbf{F}^{s}_{\text{intra}})) \odot \mathbf{F}^{s}_{\text{intra}}, \quad (3)$$

其中Fgated∈RC×HW是门控机制的输出结果，◎表示逐元素乘法，Conv（·）和 GELU（·）分别是卷积运算和 GELU 激活函数[26]。随后，我们通过特征压缩与重构对这一变换后的特征进行优化：

$$\mathbf{F}^{o}_{\text{Intra}} = \text{Conv}_{\text{up}}(\text{Conv}_{\text{down}}(\mathbf{F}_{\text{gated}})), \quad (4)$$

在哪里$\mathbf{F}^{o}_{\text{Intra}} \in \mathbb{R}^{C\times HW}$输出特征是经过调制处理后的结果，其中Convup（·）和Convdown（·）分别代表将通道维度提升至更高层级和降低至更低层级的卷积操作。该设计促使模型聚焦于数据的关键特征。

接下来，为更新各层的查询键内缓存（Intra Query-Key Cache），我们按以下方式计算每个头部的查询(Q)与键(K)投影之和：

$$F^2=Q^2+Ki, \quad (5)$$

$$\text{Fintra=Concat（}F^1\text{，…，Fh）}$$

查询-键缓存间调制与更新机制。给定两个输入张量：缓存间张量Finter∈RC'×C "和查询-键点积注意力 Fatt∈RCi×C；首先将Finter缩放为Finter∈RC:×C:，随后通过将Fatt与缩放后的特征图Finter相加计算调制分量，最终得到：

$\mathbf{F}^{s}_{\text{inter}} = \mathbf{F}_{\text{att}} + \hat{\mathbf{F}}_{\text{inter}}$.随后我们计算像素级调制的‘尺度’和‘位移’分量：

shift,

Fm=Fscale ◎Fatt+Fshift, (6)

where Fm ∈RC:×Ci is the modified feature map, Wscale and Wshift are projection matrices that learned for Fscale∈RC:×Ci and Fshift∈RC:×Ci components,respectively.Similarly,this feature is further transformed via a gating mechanism,which can be defined as:

$$\mathbf{F}^{\mathbf{o}}_{\mathbf{inter}} = \mathrm{GELU}(\mathrm{Conv}(\mathbf{F_m})) \odot \mathbf{F_m}, \tag{7}$$

where $\mathbf{F}^{\mathbf{o}}_{\mathbf{inter}} \in \mathbb{R}^{C_i \times C}$i is the output feature.

Next,to update the InterQuery-Key Cache across layers, we implement a layer-wise cache calculation process and refine the historical cache.The cache for the current layer:

$$\mathbf{F}^{l}_{\mathbf{inter}} = \sum_{i=1}^{h} \mathrm{R}(\mathbf{Q}^i {\mathbf{K}^i}^T) C_i / C, \tag{8}$$

where R(·)is a function that resizes the feature map of each head to a uniform shape,and C;/C is the assigned weight of each feature,with C being the total number of channels. Building upon the cache of the current layer,we progressively update the inter-layer cache as follows:

$$\mathbf{F_{inter}} = \alpha \mathbf{F_{inter}} + (1 - \alpha) \mathbf{F}^{l}_{\mathbf{inter}}, \tag{9}$$

where the hyper-parameter a controls the degree of information flow within the inter-layer cache.

4.Experiments

In this section,we evaluate HINT on 12 benchmark datasets across 5 typical image restoration tasks,including low-light enhancement,dehazing,and desnowing,denoising,and deraining.Due to the limited space,additional results (image deraining and denoising)and detailed experimental settings are provided in the supplementary material.

4.1.Experimental Settings

Implementation Details.HINT consists of an encoder-decoder with N_1=4 levels,where both the encoder and decoder share the same block structure:N_2=[4,6,6,6]. At the 4-th level,the encoder and decoder block are unified into a bottleneck layer,following the design of [69].The refinement stage contains N_3=4 blocks,and the embedding dimension C is set as 48.The number of attention heads is 4,with the dimensional ratio of [1,2,2,3].The reranking strategy adopted in HINT is built on Pearson correlation-based similarity.The hyper-parameter a in Equation (9)is experimentally set to 0.9.The AdamW optimizer is adopted to train HINT.The widely adopted loss functions [63,89] are employed to constrain the model training.

Metrics.We employ the popular metrics,including peak signal-to-noise ratio(PSNR)[68]and structural similarity(SSIM),to evaluate the restored result with the reference image. Besides,the non-reference metric,i.e., MANIQA [76],is adopted to measure the restoration performance on real-world inputs.In the tables,we highlight and underline the best and second-best scores,respectively.

Table 1.Quantitative comparisons on LOL-v2 [77]for low-light enhancement.*indicates unsupervised methods.

Method	LOL-v2-real PSNR SSIM	LOL-v2-syn PSNR SSIM	Average PSNR SSIM
*EnGAN[27](TIP'21)	18.230.617	16.570.734	17.40.676
*RUAS [39](CVPR' 21)	18.370.723	16.550.652	17.460.688
*QuadPrior [67](CVPR' 24)	20.480.811	16.110.758	18.300.785
KinD[87](MM'19)	14.740.641	13.290.578	14.020.610
Uformer [69](CVPR' 22)	18.820.771	19.660.871	19.240.821
Restormer [82](CVPR'22)	19.940.827	21.410.830	20.680.829
MIRNet [81](ECCV'20)	20.020.820	21.940.876	20.980.848
Sparse [77](TIP'21)	20.060.816	22.050.905	21.060.861
MambaIR[24](ECCV' 24)	21.250.831	25.550.929	23.400.880
SNR-Net [75](CVPR' 22)	21.480.849	24.140.928	22.810.889
IGDFormer [71](PR' 25)	22.730.833	25.330.937	24.030.885
Retinexformer [2](ICCV' 23)	22.800.840	25.670.930	24.240.885
MambaLLIE[72](NeurIPS'24)	22.950.847	25.870.940	24.410.894
HINT(Ours)	23.110.884	27.170.950	25.140.917

4.2.Main Results

Low-Light Enhancement.We compare the proposed HINT with state-of-the-art low-light enhancement methods on the LOL-v2 dataset in Table 1.As one can see,HINT performs favorably against considered approaches in terms of PSNR and SSIM on both real and synthetic subsets. In particular,when averaged across both subsets,HINT achieves a significant 0.9 dB improvement on PSNR over Retinexformer [2],which is the first Transformer-based pipeline designed specifically for low-light enhancement. Compared to general image restoration schemes based on diverse architectures,e.g.,CNN-based [81],Transformer-based [69,82],and Mamba-based [24],HINT receives at least 1.74 dB gain.Notably,our model yields a remarkable 1.11dB performance improvement on PSNR over the recent algorithm IGDFormer [71].The qualitative comparison is depicted in Figure 4.In the top row,previous methods fall short of recovering satisfactory results.They either meet over-/under-exposed issue [39,77](Figure 4d and 4f)or struggle to restore the true colors [81,82,87](Figure 4c,4e and 4g).For the bottom case,the image restored by HINT is closer to the reference one,where other algorithms introduce noticeable blur patterns [77,81,82,87](Figure 4c,4e, 4f,4g),or cause a under-exposed problem [39](Figure 4d).

Snow Removal.For image desnowing,we conduct experiments on the Snow100K [42]dataset.HINT archives the best PSNR score and a competitive SSIM result among all methods.To be specific,HINT provides a significant 1.64 dB boost on PSNR over the recent general restoration pipeline AST [88]. Additionally,compared to approaches [6,7,42]that are designed for desnowing,HINT showcases the superiority of better performance.Furthermore,compared to methods [36,58]that are proposed to address adverse weather conditions,our method earns consistent benefits.For general restoration schemes,including CNN-based [4,13,14,62]and Transformer-based [69,88],

其中Fm ∈RC:×Ci为修正后的特征图，Wscale和Wshift分别为针对Fscale∈RC:×Ci和Fshift∈RC:×Ci分量学习得到的投影矩阵。类似地，该特征还通过门控机制进行进一步转换，其定义可表述为：

$$\mathbf{F}^{o}_{\text{inter}} = \mathrm{GELU}(\mathrm{Conv}(\mathbf{F}_{\mathbf{m}})) \odot \mathbf{F}_{\mathbf{m}}, \tag{7}$$

在哪里$\mathbf{F}^{o}_{\text{inter}} \in \mathbb{R}^{C_i \times C_i}$为输出特征。

接下来，为实现跨层级的InterQuery-Key缓存更新，我们采用分层缓存计算机制，并优化历史缓存数据。当前层级的缓存：

$$\mathbf{F}^{l}_{\text{inter}} = \sum_{i=1}^{h} \mathrm{R}(\mathbf{Q}^{i}\mathbf{K}^{i\,T})C_i/C, \tag{8}$$

其中R（·）为将各头部特征图调整为统一形状的函数，C；/C表示各特征的权重分配值（C为通道总数）。基于当前层的缓存，我们按以下方式逐步更新层间缓存：

$$\mathbf{F}_{\text{inter}} = \alpha\mathbf{F}_{\text{inter}} + (1-\alpha)\mathbf{F}^{l}_{\text{inter}}, \tag{9}$$

其中超参数a控制层间缓存中的信息流程度。

4. 实验

本节中，我们评估了HINT在5个典型图像修复任务中的12个基准数据集上的表现，包括弱光增强、去雾、去雪、去噪和去雨。由于篇幅限制，补充材料中提供了额外结果（图像去雨和去噪）及详细的实验设置。

4.1. 实验设置

具体实现细节。HINT模型采用 N_1 =4层级的编码器-解码器架构，其编码器与解码器共享相同结构块： N_2 = [4,6,6,6]。在第四层级，编码器与解码器模块被整合为瓶颈层，该设计沿袭了文献[69]的方案。优化阶段包含 N_3 =4个结构块，嵌入维度C设为48。注意力头数量为4个，维度比例为[1,2,2,3]。HINT采用基于皮尔逊相关系数的相似度重排序策略，实验中将公式(9)中的超参数a设为0.9。训练过程中使用AdamW优化器，并采用文献[63,89]中广泛使用的损失函数来约束模型训练。

评估指标。我们采用包括峰值信噪比（PSNR）[68]和结构相似性（SSIM）在内的常用指标，以参考图像评估修复效果。此外，采用非参考指标 MANIQA [76]来衡量真实场景输入的修复性能。表格中，最佳和次优分数分别用高亮和下划线标出。

表1. LOL-v2 [77]在低光环境下的定量比较
增强.*表示无监督方法。

方法	LOL-v2-真实 PSNR SSIM	LOL-v2-syn PSNR SSIM	平均 PSNR SSIM
*恩甘[27]（TIP ‘21）	18.230.617	16.570.734	17.40.676
*RUAS [39]（CVPR ‘21）	18.370.723	16.550.652	17.460.688
*QuadPrior [67](CVPR'24)	20.480.811	16.110.758	18.300.785
KinD[87]（MM ‘19）	14.740.641	13.290.578	14.020.610
Uformer [69]（CVPR ‘22）	18.820.771	19.660.871	19.240.821
Restormer [82]（CVPR ‘22）	19.940.827	21.410.830	20.680.829
MIRNet [81](ECCV'20)	20.020.820	21.940.876	20.980.848
稀疏[77]（TIP ‘21）	20.060.816	22.050.905	21.060.861
MambaIR[24](ECCV'24)	21.250.831	25.550.929	23.400.880
SNR-Net [75]（CVPR ‘22）	21.480.849	24.140.928	22.810.889
IGDFormer [71](PR'25)	22.730.833	25.330.937	24.030.885
Retinexformer [2](ICCV'23)	22.800.840	25.670.930	24.240.885
MambaLLIE[72]（NeurIPS ‘24）	22.950.847	25.870.940	24.410.894
提示（我们的）	23.110.884	27.170.950	25.140.917

4.2. 主要结果

弱光增强技术对比。我们在表1中将提出的HINT方法与现有最先进的弱光增强方法在LOL-v2数据集上进行了对比。如表所示，HINT在真实和合成子集上的PSNR 和 SSIM 均优于其他方法。特别值得注意的是，当对两个子集进行平均计算时，HINT在 PSNR 上的表现较Retinexformer[2]（首个专为弱光增强设计的Transformer架构）实现了显著的0.9dB提升。相较于基于不同架构的通用图像修复方案（如CNN架构[81]、Transformer架构[69,82]和Mamba架构[24]），HINT至少获得1.74dB的增益。尤为突出的是，我们的模型在 PSNR 上较近期算法IGDFormer[71]实现了1.11dB的性能提升。定性对比结果如图4所示：上排数据表明，现有方法均未能获得理想效果——要么存在过曝/欠曝问题[39,77]（图4d和4f），要么难以还原真实色彩[81,82,87]（图4c、4e和4g）。在最差情况下，HINT算法恢复的图像更接近参考图像，而其他算法则会产生明显模糊纹路[77,81,82,87]（图4c、4e、4f、4g），或导致曝光不足问题[39]（图4d）。除雪处理方面，我们在Snow100K数据集[42]上进行了实验。HINT在所有方法中取得了最佳 PSNR 分数和具有竞争力的 SSIM 结果。具体而言，相较于近期通用修复流程AST[88]，HINT在 PSNR 上实现了1.64分贝的显著提升。此外，与专为除雪设计的方法[6,7,42]相比，HINT展现出更优的性能优势。再者，相较于针对恶劣天气条件提出的方法[36,58]，我们的方法持续获得显著优势。对于通用修复方案，包括
基于CNN的[4,13,14,62]和基于Transformer的[69,88]

(a)Input (b)Reference (c)KinD[87] (d)RUAS[39] (e)MIRNet [81] ()Sparse [77](g)Restormer [82](h)HINT(Ours)

Figure 4.Qualitative results on LOL-v2 [77]for low-light enhancement.The top case is from the synthetic subset,whereas the bottom one is from thereal subset.Compared to other techniques,HINT generates vivid images without introducing noticeable color distortion.

(a)Input (b)Reference (c)HDCW[7] (d)JSTASR [6] (e)Uformer [69] (f)AST[88] (g)HINT(Ours)

Figure 5.Qualitative results on Snow100K [42]for snow removal.HINT offers a clear result,while the images generated by other considered approaches remain noticeable snow artifacts.

Table 2.Quantitative results on Snow100K [42]for snow removal.

Method	JSTASR ECCV'20 [6]	All in One CVPR'20 [36]	Uformer CVPR'22 [69]	DesnowNet TIP'18 [42]	HDCW ICCV' 21 [7]	Trans Weather CVPR'22 [58]
PSNR	23.12	26.07	29.80	30.50	31.54	31.82
SSIM	0.86	0.88	0.93	0.94	0.95	0.93
Method	NAFNet ECCV'22 [4]	AST CVPR'24 [88]	FocalNet ICCV' 23 [13]	SFNet ICLR'23 [14]	ConvIR-S TPAMI'24 [62]	HINT (Ours)
PSNR	32.41	32.50	33.53	33.79	33.79	34.14
SSIM	0.95	0.96	0.95	0.95	0.95	0.94

Table 3.Quantitative comparison on SOTS [33]for haze removal.

Method	EPDN CVPR'19 [52]	FDGAN AAAI'20 [18]	AirNet CVPR'22 [34]	InstructIR ECCV' 24 [11]	Restormer CVPR'22 [82]	NAFNet ECCV' 22 [4]
PSNR	22.57	23.15	23.18	30.22	30.87	32.41
SSIM	0.863	0.921	0.900	0.959	0.969	0.970
Method,	FSNet TPAMI'24 [15]	PromptIR NeurIPS'23 [50]	DehazeFormer TIP'23 [56]	AdaIR ICLR'25 [16]	NDR-Restore TIP'24 [78]	HINT (Ours)
PSNR	31.11	31.31	31.78	31.80	31.96	32.24
SSIM	0.971	0.973	0.977	0.981	0.980	**0.981**

HINT shows at least a 0.35dB performance boost on PSNR. The visual comparisons are shown in Figure 5,where HINT restores a clear result (Figure 5g).The desnowing techniques [6,7]offer unsatisfactory outputs(Figure 5c and 5d). The results from Transformer-based approaches [69,88]exhibit noticeable snow artifacts(Figure 5e and 5f).

Haze Removal. We perform image dehazing experiments on the SOTS [33]dataset,where 11 representative approaches are considered to make a comparison in Table 3. As can be seen,HINT outperforms all other methods on both PSNR and SSIM metrics.It is worth noting that,HINT surpasses all the methods [18,52,56],which are elaborately designed for dehazing,by at least 0.46 dB on PSNR.

(a) Input (b) Reference (c) AirNet [34] (d) PromptR [50] (e) HINT

Figure 6.Qualitative results on SOTS [33]benchmark for haze removal.The image generated by HINT is closer to the reference one,compared to other methods.

When compared to image restoration pipelines (all-in-one paradigms [16,50,78] and general ones [4,11,15,82]), HINT consistently shows advantages.Figure 6 illustrates the qualitative results.In comparison,other methods introduce either color distortion issue [34](Figure 6c)or haze effect [50](Figure 6d),while HINT restores a vivid result.

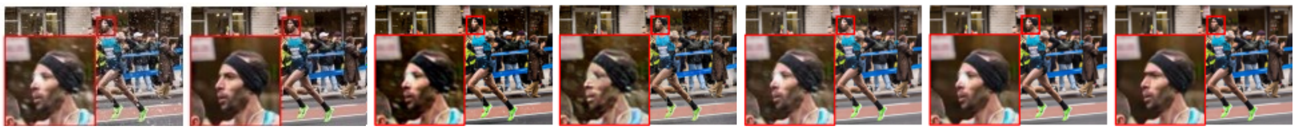
4.3.Analysis and Discussion

We have demonstrated that exploring the HMHA mechanism equipped with QKCU in a Transformer-based model brings benefits across several restoration tasks.Next,we further perform experiments to study the proposed modules and analyze the effect of each component.For ablation studies,various low-light enhancement models HINT are trained on the LOL-v2-syn subset [77].To ensure fair comparisons,all models are trained under identical experimental settings,and the FLOPs/Runtimes are computed using 256×256 input images.

Effect of HMHA.To investigate whether the proposed HMHA enhances the capability of modeling diverse contextual information,we compare it with two representative variants,including(1)Window-based Multi-head Self-

(a) (b)输入 (c)参考 (c)KinD[87] (d) RUAS [39] (e)MIRNet [81]（）稀疏 [77](g)Restormer [82](h)HINT（本研究）

图4. LOL-v2 [77]在弱光增强方面的定性结果。上图来自合成子集，下图来自真实子集。与其他技术相比，HINT生成的图像色彩鲜艳且无明显失真。



(a)输入 (b)参考(c) HDCW [7] (d) JSTASR [6] (e)Uformer[69] (f)AST[88] (g)HINT（我们的方法）图5. Snow100K[42]数据集上除雪的定性分析结果。HINT方法呈现清晰效果，而其他方法生成的图像仍存在明显积雪伪影。

表2. Snow100K[42]除雪定量结果。

方 法	JSTASR ECCV ‘20 [6]	All in One CVPR’ 20 [36]	Uformer CVPR ‘22 [69]	DesnowNet TIP’ 18 [42]	HDCW ICCV ‘21 [7]	Trans Weather CVPR’ 22 [58]
信号-噪音功率比	23.12	26.07	29.80	30.50	31.54	31.82
SSIM	0.86	0.88	0.93	0.94	0.95	0.93
方法	NAFNet ECCV ‘22 [4]	AST CVPR ‘24 [88]	局灶性网 ICCV ‘23 [13]	SF网络 ICLR ‘23 [14]	ConvIR-S TPAMI ‘24 [62]	提示（熊）
PSNR	32.41	32.50	33.53	33.79	33.79	34.14
SSIM	0.95	0.96	0.95	0.95	0.95	0.94

表3. SOTS [33]在雾霾清除方面的定量比较。

方法	EPDN CVPR ‘19 [52]	FDGAN AAAI’ 20 [18]	航空网 CVPR ‘22 [34]	InstructIR ECCV ‘24 [11]	Restormer CVPR ‘22 [82]	NAFNet ECCV ‘22 [4]
PSNR	22.57	23.15	23.18	30.22	30.87	32.41
SSIM	0.863	0.921	0.900	0.959	0.969	0.970
方法1	FSNet AMI ‘24 [15]	PromptIR NeurIPS’ 23 [50]	TP-消雾剂 TIP ‘23 [56]	rAdaIR ICLR’25 [16]	NDR-复位 TIP ‘24 [78]	提示（熊）
PSNR	31.11	31.31	31.78	31.80	31.96	32.24
SSIM	0.971	0.973	0.977	0.981	0.980	**0.981**

HINT在 PSNR 上至少显示出0.35dB的性能提升。视觉对比结果如图5所示，其中HINT恢复了清晰的结果（图5g）。去噪技术[6,7]提供的输出效果不尽如人意（图5c和5d）。基于Transformer-的方法[69,88]的结果存在明显的雪花伪影（图5e和5f）。

雾霾消除。我们在SOTS[33]数据集上进行图像去雾实验，表3中比较了11种代表性方法。可以看出，HI-NT在 PSNR 和 SSIM 指标上均优于其他所有方法。值得注意的是，HINT在 PSNR 上至少比所有专门设计用于去雾的方法[18,52,56] 高出0.46分贝。

(a) 输入 (b) 参考文献 (c) AirNet [34] (d) PromptR [50] (e) 提示

图6. SOTS[33]基准测试中雾霾去除的定性结果。与其他方法相比，HINT生成的图像更接近参考图像。

与图像修复流程（一体化范式[16,50,78]和通用流程[4,11,15,82]）相比，HINT始终展现出优势。图6展示了定性结果。相比之下，其他方法要么引入色彩失真问题[34]（图6c），要么产生雾霾效果[50]（图6 d），而HINT则能恢复生动的图像效果。

4.3. 分析与讨论

我们已经证明，在基于Transformer的模型中探索配备 QKCU 的 HMHA 机制能为多项修复任务带来优势。接下来，我们进一步开展实验以研究所提出的模块并分析各组件的影响。在消融研究中，多种低光增强模型HINT被训练在LOL-v2-syn子集[77]上。为确保公平比较，所有模型均在相同的实验设置下进行训练，且FLOPs/运行时间均使用256×256输入图像计算。

HMHA 效果 。为验证所提出的 HMHA 是否能提升对多样化文本信息建模的能力，我们将其与两种代表性变体进行对比，包括：(1)基于窗口的多头自

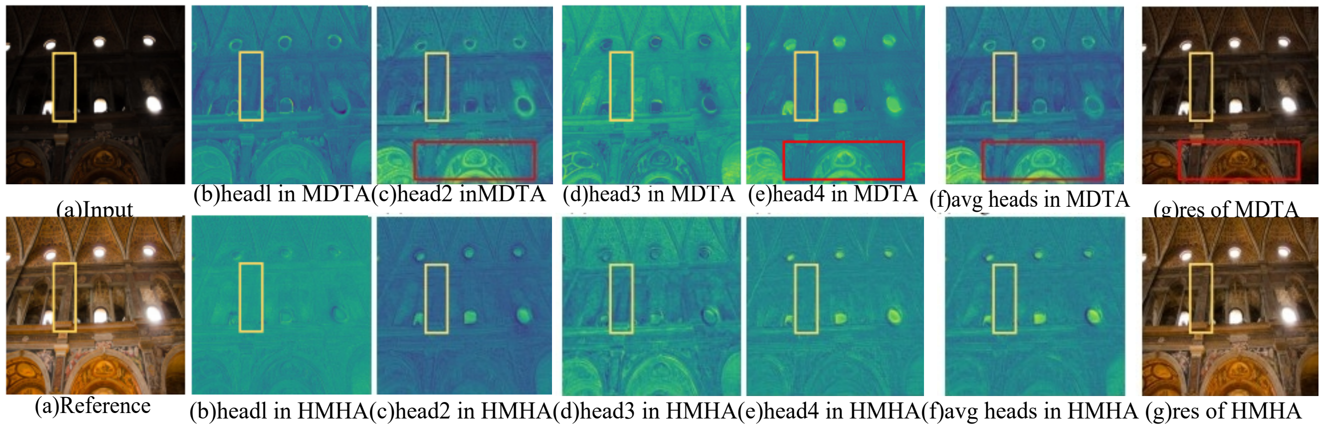


Figure 7.Feature Visualization.The top line exhibits the feature map of each head learned by MDTA,while the feature maps at the bottom ilustrate the results of HMHA.The heads in MDTA intend to focus on the same regions (red boxes),whereas the counterparts in HMHA showcase superiority in learning representations from different subspaces.As a result,the model equipped with the proposed HMHA restores a pleasant image,which is closer to the reference one(yellow boxes).

Table 4.Ablation study for different self-attention mechanisms.

Model	W-MSA [69]	MDTA [82]	HMHA Ours
PSNR/SSIM	24.19/0.941	26.42/0.948	27.17/0.950

Table 5.Ablation study of HMHA.

	Ranking Strategy	Params	PSNR	SSIM
(a)	No-Ranking [82]	24.76	26.42	0.948
(b)	Random Shuffle [84]	24.87	26.54	0.949
(c)	HMHA(Ours)	24.87	27.17	0.950

Table 6.Ablation study of QKCU.

	InraCache	InterCache	Params	PSNR	SSIM
(a)			21.34	26.47	0.949
(b)	√		23.82	26.67	0.949
(c)			22.39	26.72	0.949
(d)	√	√	24.87	27.17	0.950

Attention(W-MSA)[41],and(2)Multi-Dconv Head Transposed Attention (MDTA)[82].The quantitative comparison is reported in Table 4.The modelincorporating HMHA achieves the highest scores on both PSNR and SSIM metrics,outperforming other variants.Specifically,replacing HMHA with W-MSA or MDTA leads to significant performance drops of 2.98 dB and 0.75 dB,respectively.To investigate the effect of HMHA,we further visualize the feature maps learned by the heads in MDTA and HMHA, respectively.As shown in Figure 7,the heads in MDTA intend to focus on the same regions (indicated by red boxes), whereas the heads in HMHA showcase superiority in learning distinct representations from different subspaces.Consequently,the model with HMHA produces a result that is closer to the reference image (highlighted by yellow boxes).

We then show the effectiveness of the reranking strategy in HMHA for achieving discriminative representation

Table 7.Model efficiency analysis on LOL-V2-syn [66].

Method	IPT [3]	MIRNet [81]	Uformer [69]	Restormer [82]	HINT
FLOPs/G	6887	785	12.00	144.25	126.92
Parameters/M	115.3	31.76	5.29	26.13	24.87
Run-times/s	2.23	0.19	0.13	0.29	0.28
PSNR/dB	18.30	21.94	19.66	21.41	27.17

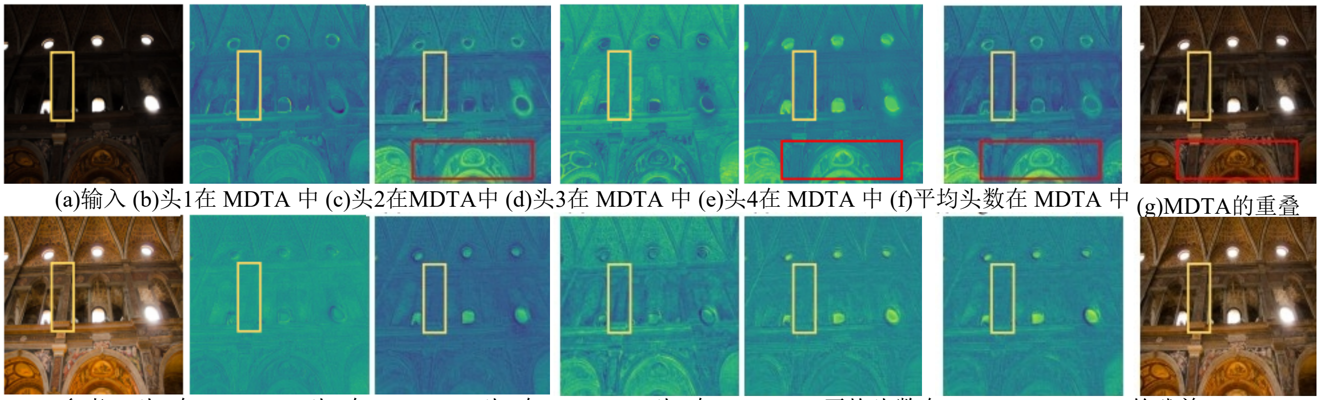
Table 8.Quantitative comparisons(MANIQA[76])on real-world datasets.These results are all obtained from testing with their best LOL-v2-syn [66]weights. [↑:Higher value denotes better quality]

Dataset	DICM [31]	MEF[46]	NPE [65]	VV[61]	Mean↑
SNR-Net [75]	0.465	0.527	0.480	0.239	0.428
Uformer [69]	0.526	0.634	0.515	0.356	0.508
Restormer [82]	0.535	0.641	0.491	0.356	0.507
Sparse [77]	0.458	0.532	0.324	0.341	0.413
HINT	0.583	0.642	0.547	0.448	0.555

in Table 5. When we replace the reranking strategy(Table 5c)with a random shuffle operation [84](Table 5b),we observe a performance drop of 0.63 dB on PSNR.In contrast,compared to the model without any ranking mechanism (Table 5a),which is degraded to vanilla MHA [82], our HMHA achieves a significant 0.75 performance boost. These results highlight the positive impact of the reranking strategy in enhancing the expressive power of the model.

Effect of QKCU.To show the effectiveness of the proposed QKCU module,we conduct experiments on different variants in Table 6.Disabling either IntraCache or InterCache within QKCU results in performance declines of 0.45 dB and 0.5 dB,respectively,indicating the importance of these modules in restoring high-quality images.In total, theQKCU mechanism brings 0.7 dB performance gain with limited computation loads(16.5%in Parameters).

Model efficiency. We provide the comparison of modelefficiency in terms of complexity(FLOPs and Parameters),



(a)输入 (b)头1在 MDTA 中 (c)头2在MDTA中 (d)头3在 MDTA 中 (e)头4在 MDTA 中 (f)平均头数在 MDTA 中 (g)MDTA的重叠

(a)参考(b)头1在 HMHA (c)头2在 HMHA (d)头3在 HMHA (e)头4在 HMHA (f)平均头数在 HMHA (g)HMHA的残差

图7.特征可视化。上排展示了 MDTA 学习到的每个头部的特征图，而下排的特征图则展示了 HMHA 的结果。 MDTA 中的头部旨在聚焦于相同的区域（红色框），而 HMHA 中的对应头部则在从不同子空间学习表征方面表现出优越性。因此，配备所提 HMHA 的模型恢复了一幅更接近参考图像（黄色框）的悦目图像。

表4. 不同自注意力机制的消融研究

模型	W-MSA [69]	人力发展与训练法 [82]	HMHA 我们的
PSNR/SSIM	24.19/0.941	26.42/0.948	27.17/0.950

表5. HMHA 消融研究

排名策略	参数	信号-噪音功率比	SSIM
(a) 无排名[82]	24.76	26.42	0.948
(b) 随机洗牌[84]	24.87	26.54	0.949
(c) HMHA（我们的）	24.87	27.17	0.950

表6. QKCU 消融研究

	Inra缓存	互连高速缓存	参数	信号-噪音功率比	SSIM
(a)			21.34	26.47	0.949
(b)	√		23.82	26.67	0.949
(c)			22.39	26.72	0.949
(d)	√	√	24.87	27.17	0.950

表7. LOL-V2-syn模型效率分析[66]。

方法	IPT [3]	MIRNet [81]	Uformer [69]	Restormer [82]	HINT
浮点运算次数/千次	6887	785	**12.00**	144.25	126.92
参数/测量	115.3	31.76	**5.29**	26.13	24.87
运行时间/s	2.23	0.19	**0.13**	0.29	0.28
PSNR/dB	18.30	21.94	**19.66**	21.41	**27.17**

表8.真实世界数据集的定量比较（MANIQA [76]）。所有结果均通过使用其最佳LOL-v2-syn [66]权重进行测试获得。[个 :数值越高表示质量越好]

数据集	DICM [31]	MEF[46]	NPE [65]	VV[61]	Mean个
SNR-Net [75]	0.465	0.527	0.480	0.239	0.428
Uformer [69]	0.526	0.634	0.515	0.356	0.508
Restormer [82]	0.535	0.641	0.491	0.356	0.507
稀疏[77]	0.458	0.532	0.324	0.341	0.413
提示	**0.583**	**0.642**	**0.547**	**0.448**	**0.555**

注意力（W-MSA）[41]，以及(2)多Dconv头转置注意力（MDTA）[82]。定量比较结果见表4。采用 HMHA 的模型在 PSNR 和 SSIM 指标上均取得最高分，优于其他变体。具体而言，用W-MSA或 MDTA 替代 HMHA 会导致性能分别下降2.98 dB和0.75 dB。为探究 HMHA 的影响，我们进一步可视化了 MDTA 和 HMHA 中头部学习到的特征图，分别。如图7所示，MDTA 中的头部倾向于聚焦相同区域（红色框标示），而 HMHA 中的头部在学习不同子空间的独立表征方面更具优势。因此，采用 HMHA 的模型生成的结果更接近参考图像（黄色框标示）。

随后我们展示了 HMHA 中重排序策略在实现判别性表征方面的有效性

在表5中，当我们用随机洗牌操作[84]（表5 b）替换重排序策略（表5 c）时，观察到 PSNR 性能下降了0.63分贝。相比之下，与未采用任何排序机制的模型（表5a）相比——该模型退化为基础MHA[82]——我们的 HMHA 实现了显著的0.75分贝性能提升。这些结果凸显了重排序策略在增强模型表达能力方面的积极影响。

QKCU 效果。为验证所提出的 QKCU 模块的有效性，我们在表6中对不同变体进行实验。禁用 QKCU 中的Intra-Cache或InterCache会导致性能分别下降0.45dB和0.5 dB，这表明这些模块对恢复高质量图像的重要性。总体而言，QKCU机制在有限计算负载（参数量减少16.5%）下实现了0.7dB的性能提升。

模型效率。 我们从复杂度（浮点运算次数和参数量）方面对模型效率进行比较，

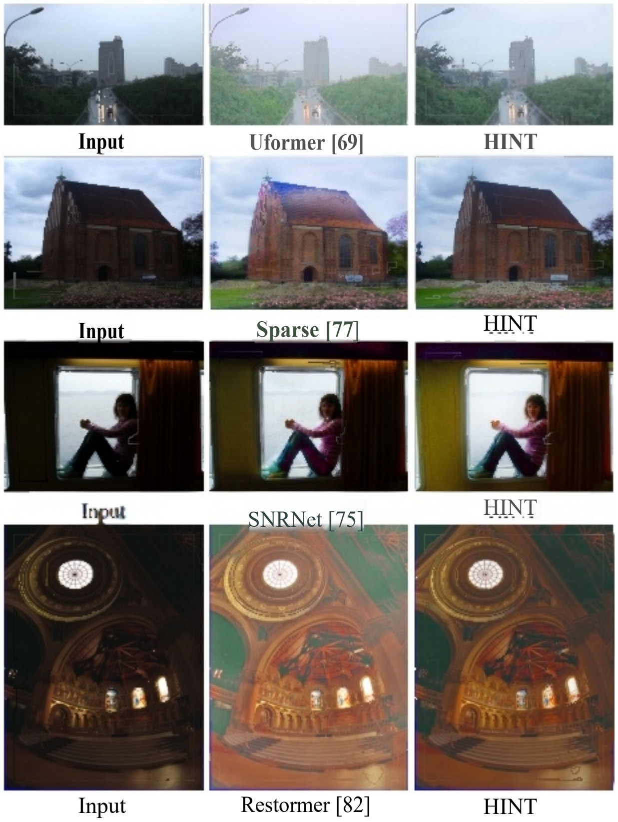


Figure 8.Visualizations on the real-world benchmarks,including NPE [65],DICM[31],VV [61],and MEF [46](top to bottom). HINT restores a pleasant result,whereas the considered technologies meet under-/over-exposed problem [75,82],or trigger color distortion [77],or remain significant artifacts [69].

latency(Run-times),and performance(PSNR),as shown in Table 7. HINT obtains the highest PSNR score,while maintaining lower model complexity than CNN-based MIRNet [81],Transformer-based IPT [3]and Restormer [82].

Evaluation on real-world scenarios. To assess the performance of HINT under real-world scenarios,we test it on real-world dataset without ground-truth,including DICM[31],MEF[46],NPE[65],and VV[61].The quantitative comparisons are summarized in Table 8,where HINT outperforms all other methods.Furthermore,as shown in Figure 8,HINT restores a visually pleasing restoration, whereas the considered technologies meet under-/over-exposed problem [75,82],trigger color distortion [77],or remain significant artifacts [69].

Application to high-level vision Task. We conduct the low-light object detection experiment on the ExDark [44] dataset.HINT is pre-trained on the LOL-v2-syn subset and directly applied to enhance the low-light images,with the YOLO-v3[53]model serving as the detector.The enhanced images bring benefits for the downstream task in qualitative (Figure 9)and quantitative ways(Table 9).

Table 9.Quantitative result of low-light object detection Ex-Dark [44]benchmark.HINT positively impacts the downstream task,and boosts the average precision(AP)scores to 7.6%.

Metric	Input	HINT	△
Mean(average precision,AP)	45.1	52.7	+7.6

Figure 9.Visual comparison of low-light object detection.Compared to the low-light input (left),the detector can predict a well-placed bounding box on the image restored by HINT(right).

Figure 10.Failure Case.HINT,which is pre-trained on the synthetic dataset,meets the challenges of restoring input under extremely low-light conditions.

5.Conclusion

We present HINT,a Transformer-based pipeline for image restoration.Without bells and whistles,HINT is simple yet effective for 5 typical restoration tasks on **12** benchmarks, which demonstrates the effect of two design choices:1) Hierarchical Multi-Head Attention(HMHA);2)Query-Key Cache Updating(QKCU).By leveraging HMHA to migrate redundancy that is rooted in the vanilla Multi-Head Attention(MHA),and introducing QKCU to enhance interactions between heads via intra-and inter-modulation,HINT is able to perform favorably against previous state-of-the-art algorithms in terms of model complexity and accuracy. To the best of our knowledge,HINT is the first work to restore high-quality images by exploring an efficient MHA mechanism within the widely-used Transformer architecture.This work provides a promising direction to achieve superior restoration performance,and we hope this community can ignite the interest and benefit from it.

Limitations. While HINT offers a plausible solution for image restoration,an interesting way to achieve better results is to solve the failure case it triggers.As shown in Figure 10,HINT struggles to recover some cases under extremely low-light conditions.One potential reason for this could be the domaingap between the synthetic datasets used for pre-training and real-world data.Collecting large-scale real-world datasets for further training could be a valuable direction for future work.

图8.真实世界基准的可视化结果，包括 NPE [65]、 DICM [31]、VV[61]和MEF[46]（从上到下）。HINT恢复了令人满意的结果，而所考虑的技术则存在曝光不足/过度的问题[75,82]，或引发色彩失真[77]，或残留显著伪影[69]。

延迟（运行时间）和性能（PSNR），如表7所示。HINT获得最高的 PSNR 分数，同时保持比基于CNN的MIR Net[81]、基于Transformer的 IPT [3]和Restormer[82]更低的模型复杂度。

真实场景评估。为评估HINT在真实场景下的性能，我们在无真实标签的真实数据集上进行测试，包括DICM [31]、MEF[46]、NPE [65]以及 VV[61]. 该量表8总结了对比结果，其中HINT优于所有其他方法。此外，如图8所示，HINT能恢复视觉效果良好的图像，而所考虑的技术则存在欠曝/过曝问题[75,82]、触发色彩失真[77]或残留显著伪影[69]。

应用于高级视觉任务。我们在ExDark[44]数据集上进行了弱光目标检测实验。HINT在LOL-v2-syn子集上进行预训练，并直接应用于增强弱光图像，以YOLO-v3 [53]模型作为检测器。增强后的图像在定性（图9）和定量方式（表9）上为下游任务带来了益处。

表9. 低光物体检测Ex Dark[44]基准的定量结果。HINT对下游任务产生积极影响，将平均精度（AP）分数提升至7.6%。

指标	输入	提示	△
平均值（平均数） 精确度，AP）	45.1	52. 7	+7. 6

图9. 低光环境物体检测的视觉对比。与低光输入（左）相比，该检测器能够对HINT恢复的图像（右）预测出位置准确的边界框。

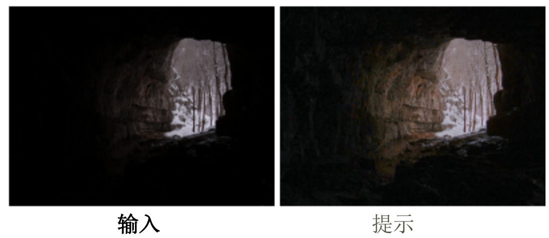


图10. 失败案例。HINT在合成数据集上进行预训练，能够应对极低光照条件下输入恢复的挑战。

5. 结论

我们推出HINT——一个基于Transformer架构的图像修复流程。该方案摒弃了花里胡哨的装饰，仅通过简单有效的方式，在**12**个基准测试中完成了5项典型修复任务，充分展现了两大设计选择的效果：1）分层多头注意力机制（HMHA）；2）查询键缓存更新（QKCU）。通过利用 HMHA 迁移源自基础多头注意力（MHA）的冗余信息，并引入 QKCU 增强头间通过模内与模间调制的交互作用，HINT在模型复杂度与准确度方面 优于现有主流算法 - 的 - 该 领域算法。据我们所知，HINT是首个在广泛使用的Transformer架构中探索高效MHA机制以重建高质量图像的研究成果。这项工作为实现卓越修复性能指明了方向，我们期待该成果能激发学界兴趣并带来实际效益。

局限性。虽然HINT为图像修复提供了一个可行的解决方案，但 一个提升效果的有趣方法是解决其引发的故障场景 。 如图 10 所示，HINT在极低光照条件下难以恢复某些案例。这可能源于预训练使用的合成数据集与真实世界数据之间的领域差异。收集大规模真实世界数据集用于后续训练，可能成为 未来研究的重要方向 。